RiskLens

Fixed-Income Intelligence, Portfolio Optimisation & Risk Analytics Platform

India-focused MVP · Project Whitepaper

Product	RiskLens — fixed-income analytics & decision-support
Market	India (extensible to other markets)
Users	Individual investors & investment professionals
Stack	Next.js · FastAPI · SQLAlchemy · PostgreSQL · NumPy/SciPy
Repository	github.com/gowtham9701/RiskLens
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Disclaimer. All bundled data is SIMULATED. RiskLens is an analytics and decision-support tool. It does not provide investment advice and does not guarantee returns, capital protection, liquidity or credit quality. Yields, liquidity scores and stress results are indicative estimates only.

1. Executive Summary

RiskLens is a full-stack platform that helps individuals and professionals discover, value, construct, optimise and monitor fixed-income portfolios across Indian instrument classes — Government Securities, Treasury Bills, State Development Loans, corporate bonds/NCDs, bank fixed deposits and debt mutual funds. It pairs an institutional-grade, independently tested financial engine with a transparent, never-fabricate-data philosophy: every figure is traceable to a source and timestamp, and nothing is presented as a guarantee.

The platform answers the core investor question — *is the extra yield worth the risk, and how do I build and watch a portfolio around it?* — through risk-adjusted analytics (duration, convexity, DV01), a transparent 0–100 liquidity score, a cash-flow-weighted portfolio yield, a SciPy constrained optimiser, scenario stress testing, and proactive alerts.

2. Business Idea

India's fixed-income market is large and increasingly accessible to retail investors via RBI Retail Direct, online bond platforms and debt mutual funds. The tools to *evaluate* these instruments, however, remain fragmented, spreadsheet-driven and skewed toward institutions. RiskLens provides a single, transparent intelligence layer positioned explicitly as analytics and decision-support — not a broker and not an advisor.

Value proposition

- **Risk-adjusted transparency** — every yield is shown next to its credit, liquidity and interest-rate risk, with the methodology exposed.
- **Correct financial maths** — cash-flow-weighted portfolio yield and instrument-specific methods, avoiding the common errors in consumer tools.
- **Actionable construction & monitoring** — constraint-based portfolio building, stress testing, cash-flow laddering and event alerts in one place.
- **Extensible & provider-agnostic** — data adapters can be swapped without touching the calculation engine; new instrument classes and markets can be added.

3. Problem Statement

Challenge	Why it hurts investors
Yield ≠ safety	Higher coupons often hide credit, liquidity or duration risk retail tools never surface.
Fragmented data	Prices, ratings, secondary-market liquidity and issuer news live in separate silos.
Opaque liquidity	Investors confuse large issue size with the real ability to sell; many bonds barely trade.
Wrong maths	Averaging individual YTMs is incorrect; bond formulas are wrongly applied to FDs/funds.
No portfolio view	Little support for constrained construction, laddering, stress testing or downgrade impact.
Reactive monitoring	Downgrades and liquidity deterioration are discovered too late.

4. Solution Provided by RiskLens

Problem	RiskLens solution
Yield ≠ safety	Duration, convexity, DV01, credit & liquidity scoring shown beside every yield.
Fragmented data	One data model unifying instruments, prices, ratings, liquidity, news & issuers.
Opaque liquidity	Transparent 0–100 liquidity score (5 weighted components) + estimated liquidation days/cost.
Wrong maths	Cash-flow-weighted portfolio YTM (IRR); instrument-specific methods for FDs & funds.
No portfolio view	SciPy constrained optimiser, maturity ladder, stress testing, cash-flow calendar.
Reactive monitoring	Alerts for downgrades, negative outlooks, liquidity drops, concentration & maturities.

Core principle: maximise income/yield *subject to transparent risk constraints*, and always disclose data sources, timestamps, assumptions and limitations.

5. Target Users

Individual investors

- Enter budget, horizon and income needs; receive a clear proposed portfolio.
- Compare alternatives, track income and maturities, and receive relevant alerts.
- Interface understandable without professional financial knowledge.

Investment professionals

- Detailed bond valuation, portfolio YTM, duration/convexity/DV01, rating migration.
- Historical liquidity analytics, stress testing, data-quality monitoring and audit trails.
- Multiple portfolios, performance attribution, and Excel/PDF reporting.

6. System Architecture

RiskLens is a modular monorepo with a strict separation between the financial calculation engine, the application backend, data ingestion and the UI. The engine is a standalone, independently unit-tested Python package with no web or database dependencies, ensuring the maths is correct regardless of data provider.

Layer	Technology	Responsibility
Frontend	Next.js, TypeScript, Tailwind, Plotly	Dashboards, explore, builder, risk, reports
Backend	FastAPI, Pydantic, SQLAlchemy, JWT	REST API, auth, user-scoped portfolios
Analytics	NumPy, SciPy, pandas	Pricing, yield, risk, liquidity, optimiser, stress
Database	PostgreSQL (29 tables, NUMERIC)	Normalised, precise, user-isolated storage
Reporting	OpenPyXL, ReportLab	Excel & PDF portfolio reports
Deploy	Docker, Render, Vercel, Neon, CI	Reproducible build & cloud deployment

7. Application Modules

- **Executive Dashboard** — value, gain/loss, weighted yield, income, duration, DV01, allocations, maturity ladder.
- **Instrument Database** — master records with full terms, prices, ratings, day-count schedules.
- **Calculation Engine** — valuation, cash-flow generation, yield-curve shifts.
- **Credit Intelligence** — rating history, outlooks, downgrade alerts with portfolio impact.
- **Liquidity Intelligence** — volume analysis, spread/depth tracking, illiquidity flags.
- **News & Events** — issuer/instrument-linked disclosures with summaries (LLM-ready).
- **Portfolio Builder** — questionnaire + SciPy optimiser producing compliant target portfolios.
- **Risk & Stress Testing** — rate/spread shifts, downgrades, default-recovery, liquidity shocks.
- **Cash-Flow Calendar** — coupons, FD interest, maturities, reinvestment needs.
- **Portfolio Management** — multi-portfolio, transactions, CSV import, attribution.
- **Alerts** — downgrades, negative outlooks, liquidity drops, maturity & concentration risks.
- **Reporting & Data Quality** — PDF/Excel export + exception tracking console.

8. Financial Methodology (highlights)

- **Day counts**: ACT/365F, ACT/360, ACT/ACT, 30/360, 30E/360.
- **Pricing & yield**: present value, clean/dirty price, YTM/YTC/YTP (Brent solver), zero-coupon & current yield.
- **Risk**: Macaulay & modified duration, convexity, DV01, effective duration; first+second order price change.
- **Portfolio**: cash-flow-weighted YTM (IRR of aggregated flows) — never a naive average; XIRR; weighted duration/DV01.
- **Liquidity score**: ADTV 25%, trading frequency 20%, spread 20%, depth 20%, recency 15%; 'Insufficient data' when sparse.
- **Optimiser**: SciPy SLSQP; hard constraints on credit, liquidity, duration, issuer/sector caps, ticket size; reports infeasibility.
- **Stress**: ±50/±100 bps rate & spread shocks, downgrade, default-with-recovery, volume drop, forced liquidation.

Validated by 30 analytics unit tests (par/premium/discount pricing, YTM round-trips, zero-coupon duration = maturity, DV01 ≈ ModDur·P·1bp, optimiser constraints) and 15 API integration tests. All monetary/yield values are stored as SQL NUMERIC, never float.

9. Data Integrity, Security & Controls

- Provider-agnostic adapters; synthetic-data mode by default; manual CSV import path.
- Data-quality validations: duplicates, missing/stale prices, invalid maturities, allocation totals.
- bcrypt password hashing, JWT auth, strict per-user portfolio isolation, parameterised ORM queries.
- Secrets only in environment variables; audit-log and data-exception tables for traceability.
- Graceful degradation: a news/LLM outage never blocks portfolio calculations.

10. End-to-End Project Guide

Local run (no Docker)

Step	Command
Create venv	python3 -m venv .venv && source .venv/bin/activate
Install engine	pip install -e ./analytics
Install backend	pip install -r backend/requirements.txt
Configure	cp .env.example .env (set DATABASE_URL or use SQLite)
Seed data	python -m backend.app.seed
Run API	uvicorn backend.app.main:app --reload --port 8000
Run UI	cd frontend && npm install && npm run dev
Run tests	pytest analytics/tests backend/tests -q

Demo login: **demo@risklens.sim / demo1234**. API docs at /docs.

Cloud deployment

- **Database**: Neon (or Supabase / Render Postgres) — copy the connection string.

- **Backend**: Render Blueprint (render.yaml) — set DATABASE_URL & CORS_ORIGINS; seeds only if empty.
- **Frontend**: Vercel — Root Directory = frontend, set NEXT_PUBLIC_API_URL to the backend URL.
- **Alternatives**: Railway, Fly.io, Google Cloud Run, Koyeb (backend); Netlify, Cloudflare Pages (frontend); Coolify/Dokku/CapRover (self-host).

11. Roadmap & Known Limitations

- All bundled data is simulated; live market/rating/news feeds require licences & credentials.
- QuantLib, option-adjusted (OAS) pricing, key-rate durations and TWR are future enhancements.
- Performance: add query-batching + Redis caching and background ingestion for scale.
- Hardening: refresh-token rotation, full rate-limiting, and Alembic migrations before launch.
- Regulatory review (securities, investment-advice, data-licensing, privacy) required pre-commercial.

12. Legal & Financial Disclaimer

RiskLens is an investment analytics and decision-support platform. It must not be used as a substitute for professional investment advice. It does not guarantee returns or capital protection, does not treat ratings or secured status as risk-free, and presents estimated liquidity and indicative yields as estimates, not guarantees. All bundled data is simulated and clearly labelled. Review applicable Indian securities, investment-advice, data-licensing, privacy and consumer-protection requirements before any commercial use.